ALADDIN ISMAEL

Software Engineer
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EXPERIENCE

Application Engineer | Qcify — Nov 2024 - Oct 2026

Qcify develops computer vision quality control and optical sorting systems for agricultural manufacturing.

- **Developed and retrained PyTorch CNN models for production defect detection** with DVC, Azure Pipelines, Azure Blob Storage, and Azure DevOps, adapting models and post-processing logic to each customer's quality requirements.
- **Improved average macro F1 score from approximately 93% to 96%** through targeted image sampling and machine-specific data augmentation; standardized the evaluation metrics around domain-specific defect requirements.
- **Led the transition from multiclass to multilabel image classification** across production models and datasets, enabling detection of multiple defects per item and improving model performance on co-occurring defect categories.
- **Owned customer-facing AI engineering across Central California** for two harvest seasons, working on site with plant teams and machine operators, translating their requirements into model updates validated on live production lines.
- **Built internal Python debugging tools that reduced development iteration time by 20%**, isolating whether misclassifications came from camera artifacts or poor calibration, CNN prediction errors, or post-processing bugs.

Founding Software Engineer | DevSpot — Sep 2024 - Aug 2025

- **Owned DevOps for a hackathon platform** on Vercel and Supabase, developed by a nine-person team, now at 9,000 registered users across 15 hackathons, including events for Coinbase, Protocol Labs, and Funding the Commons.
- **Implemented multi-tenancy that gave each hackathon event its own subdomain** through middleware routing and database row-level security, after standing up the platform's Supabase PostgreSQL database and authentication.
- **Created a staging environment for QA testing** ahead of production releases and handled all database migrations.

Computer Vision Application Engineer | Wave Spectrum — Mar 2024 - Nov 2024

- **Improved production YOLOv8 object-detection models** for real-time aircraft parking assistance in hangars, analyzing false-alert statistics, identifying annotation issues, and routing failure cases into the annotation workflow.
- **Built OpenCV tooling to automate 3D boundary-point collection** for onboarding new hangars, and performed on-site calibration of hangar boundaries using physical reference points so alarms trigger reliably in real-world conditions.

Full Stack Developer | IhsanS LLC — Mar 2023 - Mar 2024

- **Built and deployed a HIPAA-compliant electronic medical records application** for a private orthopedic medical practice using React, Node.js, Express, and MongoDB, reducing the practice's patient case evaluation time by 40%.
- **Implemented role-based API authorization with Auth0 JWT**, CI/CD pipelines, and application/audit logging with Winston and AWS CloudWatch; containerized and deployed the backend services using Docker Compose and AWS EC2.
- **Created a reusable React Hook Form input library** with recursive rendering, cutting form digitization time in half.

Mobile Application Developer | Cal Poly San Luis Obispo, Student Research — Jun 2022 - Jun 2023

- **Built a React Native dashboard for agricultural engineering students and professors** to visualize tractor efficiency metrics from uploaded CAN sensor telemetry, integrating an IoT logging device with AWS S3, Lambda, and InfluxDB.

Software Engineer Intern | PilotCity — Jun 2021 - Sep 2021

- **Developed a Flutter mobile application** for a nonprofit delivering food to those in need, matching volunteer drivers with donors by location and availability and enabling live pickup tracking through the Firebase Realtime Database.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Node.js, Next.js, React, React Native, Express
ML & Computer Vision: PyTorch, TorchVision, OpenCV, YOLOv8, CNNs, multilabel classification, data augmentation, DVC
Cloud & MLOps: Docker, CI/CD, Azure Pipelines, Azure DevOps, Azure Blob Storage, AWS EC2, S3, Lambda, CloudWatch
Backend, Data & Mobile: REST APIs, PostgreSQL, MongoDB, Supabase, Firebase, InfluxDB, Flutter

EDUCATION

B.S. Computer Science | California Polytechnic State University, San Luis Obispo — June 2023